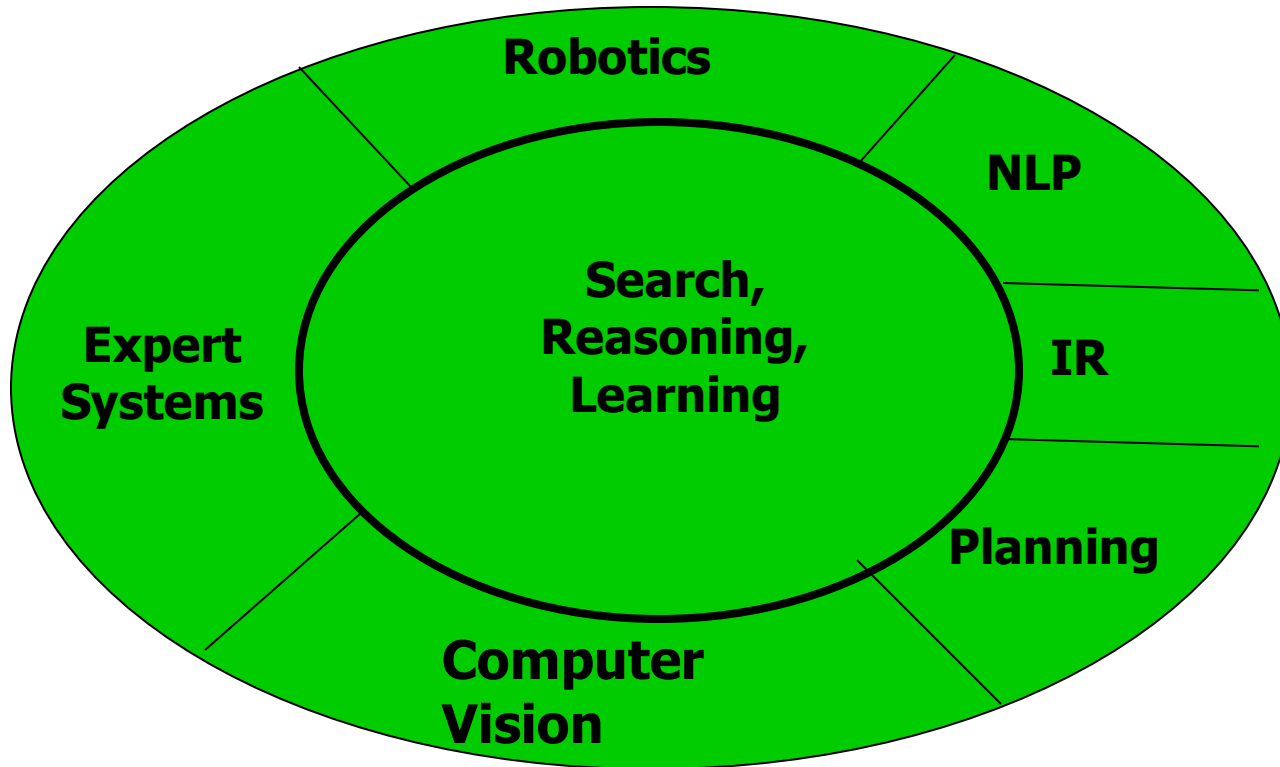


CS564: Foundations of ML

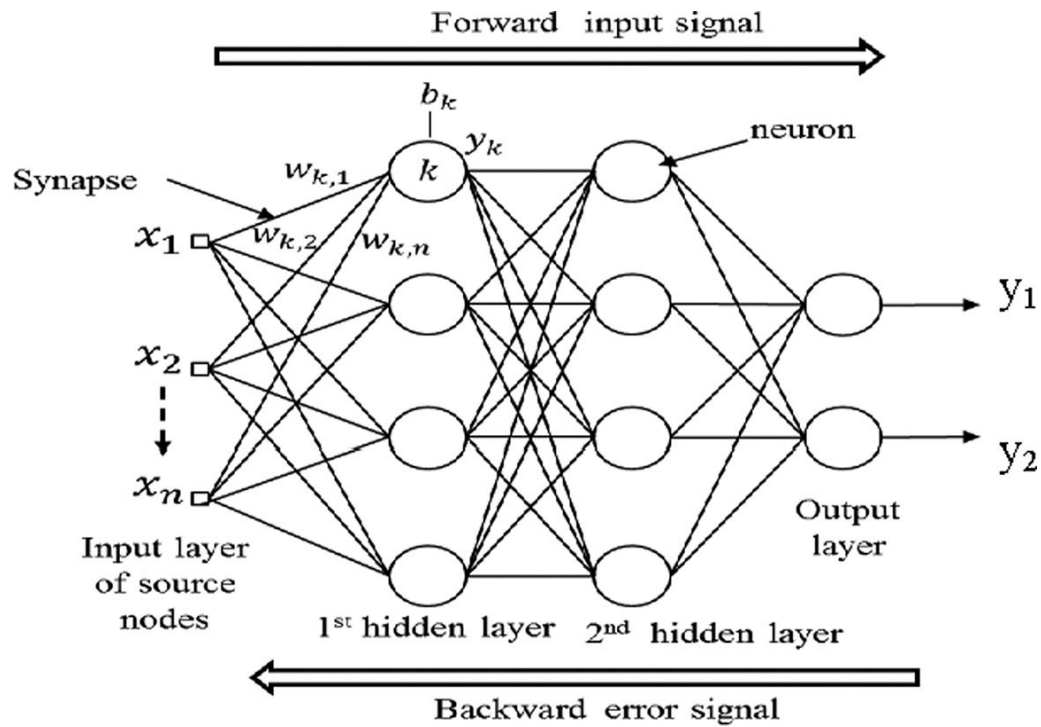
Asif Ekbal
Department of CSE

Lecture: Sequence Learning- Hidden Markov Model

AI Perspective (post-web)



ANN





Sequence Learning

Observations leading to why probability is needed

- Many intelligence tasks are sequence labeling tasks
- Tasks carried out in layers
- Within a layer, there are limited windows of information
- This naturally calls for strategies for dealing with uncertainty
- Probability and Markov process give a way

“I went with my friend to the bank to withdraw some money, but was disappointed to find it closed”

POS	Bank (N/V)	closed (V/ adj)
Sense	Bank (financial institution)	withdraw(take away)
Pronoun drop	But I/friend/money/bank	was disappointed
SCOPE	With	my friend
Co-referencing	It -> bank	

Different Models for Sequence Learning

- **Probabilistic Models**

- Hidden Markov Model (HMM)
- Maximum Entropy Markov Models
- Conditional Random Fields

- **Neural Network based Models**

- Recurrent Neural Network
- Long Short Term Memory
- Gated Recurrent Unit

Hidden Markov Models

- **Set of states:** $\{s_1, s_2, \dots, s_N\}$

- Process moves from one state to another generating a sequence of states :

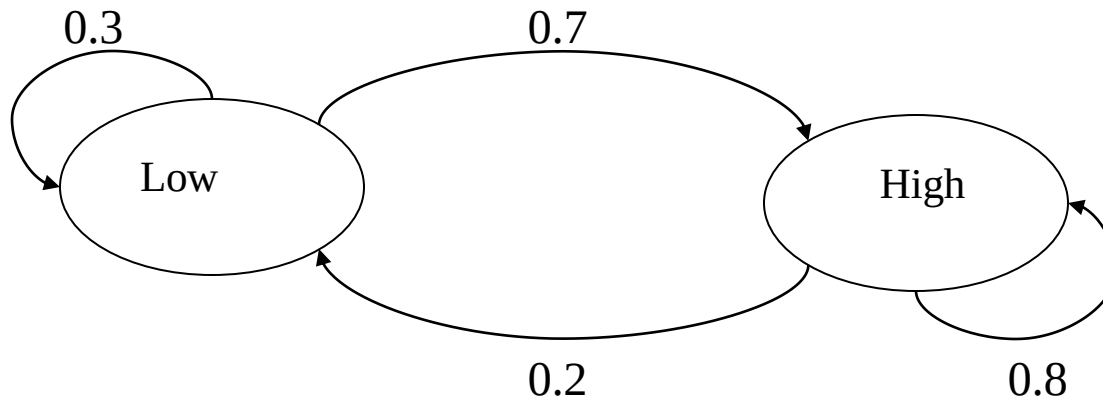
$$s_{i1}, s_{i2}, \dots, s_{ik}, \dots$$

- **Markov chain property:** probability of each subsequent state depends only on what was the previous state:

$$P(s_{ik} \mid s_{i1}, s_{i2}, \dots, s_{ik-1}) = P(s_{ik} \mid s_{ik-1})$$

- States are not visible, but each state randomly generates one of M observations (or visible states) $\{v_1, v_2, \dots, v_M\}$

Example of Markov Model



- **Two states** : 'Low' and 'High'
- **Transition probabilities**: $P(\text{'Low'} | \text{'Low'})=0.3$, $P(\text{'High'} | \text{'Low'})=0.7$, $P(\text{'Low'} | \text{'High'})=0.2$, $P(\text{'High'} | \text{'High'})=0.8$
- **Initial probabilities**: say $P(\text{'Low'})=0.4$, $P(\text{'High'})=0.6$

Computing Sequence Probability

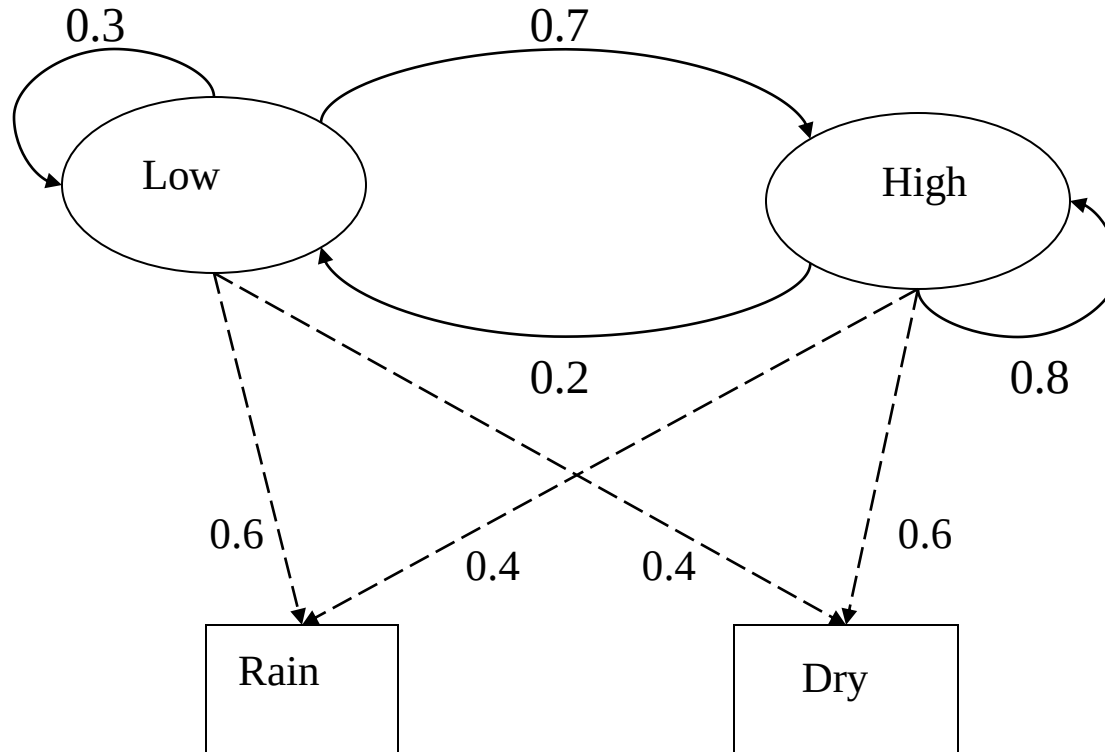
- By Markov chain property, probability of state sequence can be found by the formula:

$$\begin{aligned} P(s_{i1}, s_{i2}, \dots, s_{ik}) &= P(s_{ik} \mid s_{i1}, s_{i2}, \dots, s_{ik-1}) P(s_{i1}, s_{i2}, \dots, s_{ik-1}) \\ &= P(s_{ik} \mid s_{ik-1}) P(s_{i1}, s_{i2}, \dots, s_{ik-1}) = \dots \\ &= P(s_{ik} \mid s_{ik-1}) P(s_{ik-1} \mid s_{ik-2}) \dots P(s_{i2} \mid s_{i1}) P(s_{i1}) \end{aligned}$$

- Suppose we want to calculate a probability of a sequence of states in our example, {'Low','Low','High','High'}.

$$\begin{aligned} P(\text{'Low','Low','High','High'}) &= \\ P(\text{'Low'} \mid \text{'Low'}) P(\text{'Low'} \mid \text{'High'}) P(\text{'High'} \mid \text{'High'}) P(\text{'High'}) &= \\ = 0.3 * 0.2 * 0.8 * 0.6 \end{aligned}$$

An Example



Example (Contd.)

- **Two states** : 'Low' and 'High' atmospheric pressure
- **Two observations** : 'Rain' and 'Dry'
- **Transition probabilities**:
 $P(\text{'Low'} | \text{'Low'})=0.3$, $P(\text{'High'} | \text{'Low'})=0.7$,
 $P(\text{'Low'} | \text{'High'})=0.2$, $P(\text{'High'} | \text{'High'})=0.8$
- **Observation probabilities**:
 $P(\text{'Rain'} | \text{'Low'})=0.6$, $P(\text{'Dry'} | \text{'Low'})=0.4$,
 $P(\text{'Rain'} | \text{'High'})=0.4$, $P(\text{'Dry'} | \text{'High'})=0.36$
- **Initial probabilities**: say $P(\text{'Low'})=0.4$, $P(\text{'High'})=0.6$

Computing Observation Sequence Probability

- Suppose we want to calculate a probability of a sequence of observations in our example, {'Dry','Rain'}

- Consider all possible hidden state sequences:

$$P(\{\text{'Dry'}, \text{'Rain'}\}) = P(\{\text{'Dry'}, \text{'Rain'}\}, \{\text{'Low'}, \text{'Low'}\}) + P(\{\text{'Dry'}, \text{'Rain'}\}, \{\text{'Low'}, \text{'High'}\}) + P(\{\text{'Dry'}, \text{'Rain'}\}, \{\text{'High'}, \text{'Low'}\}) + P(\{\text{'Dry'}, \text{'Rain'}\}, \{\text{'High'}, \text{'High'}\})$$

where first term is :

$$\begin{aligned} &P(\{\text{'Dry'}, \text{'Rain'}\}, \{\text{'Low'}, \text{'Low'}\}) = \\ &P(\{\text{'Dry'}, \text{'Rain'}\} \mid \{\text{'Low'}, \text{'Low'}\}) P(\{\text{'Low'}, \text{'Low'}\}) = \\ &P(\text{'Dry'} \mid \text{'Low'}) P(\text{'Rain'} \mid \text{'Low'}) P(\text{'Low'}) P(\text{'Low'} \mid \text{'Low'}) \\ &= 0.4 * 0.4 * 0.6 * 0.4 * 0.3 \end{aligned}$$

Hidden Markov Models: *Three Important Components*

To define hidden Markov model, the following probabilities have to be specified:

- **Matrix of transition probabilities**

$$A=(a_{ij}), a_{ij}= P(s_i | s_j)$$

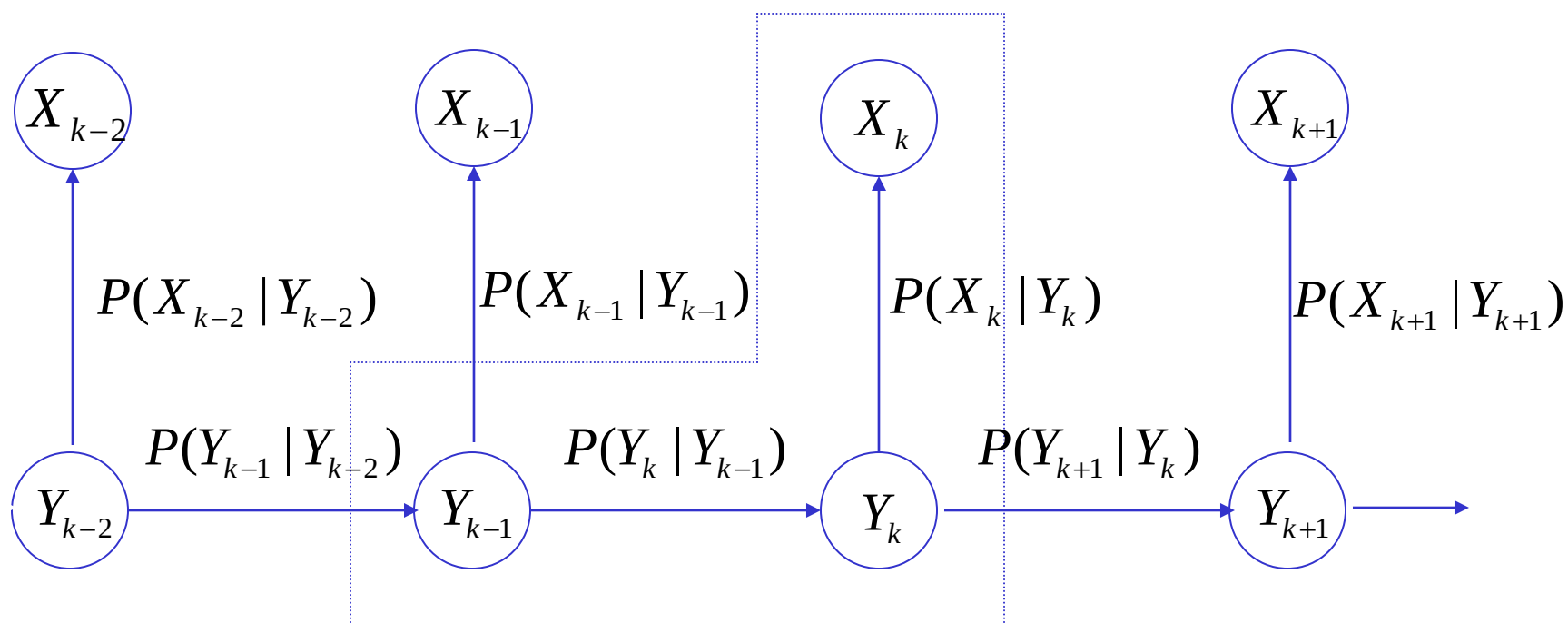
- **Matrix of observation probabilities**

$$B=(b_i(v_m)), b_i(v_m) = P(v_m | s_i)$$

- **Vector of initial probabilities** $\pi=(\pi_i), \pi_i = P(s_i)$

Model is represented by $M=(A, B, \pi)$

Dependency (1st order)



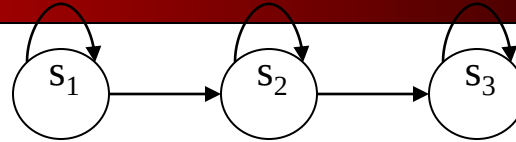
Main issues using HMMs

- **Evaluation problem:** Given the HMM $M=(A, B, \pi)$ and the observation sequence $O=o_1 o_2 \dots o_K$, calculate the probability that model M has generated sequence O
- **Decoding problem:** Given the HMM $M=(A, B, \pi)$ and the observation sequence $O=o_1 o_2 \dots o_K$, calculate the most likely sequence of hidden states s_i that produced this observation sequence O
- **Learning problem:** Given some training observation sequences $O=o_1 o_2 \dots o_K$ and general structure of HMM (numbers of hidden and visible states), determine HMM parameters $M=(A, B, \pi)$ that best fit training data

$O=o_1 \dots o_K$ denotes a sequence of observations $o_k \in \{v_1, \dots, v_M\}$.

Exercise: Character recognition with HMM(1/3)

- The structure of hidden states:

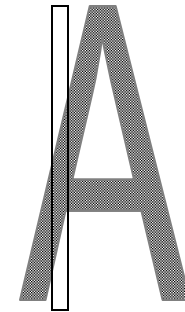


- Observation = number of islands in the vertical slice

- HMM for character 'A':

Transition probabilities: $\{a_{ij}\} = \begin{pmatrix} .8 & .2 & 0 \\ 0 & .8 & .2 \\ 0 & 0 & 1 \end{pmatrix}$

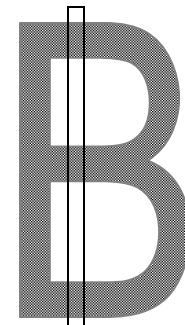
Observation probabilities: $\{b_{jk}\} = \begin{pmatrix} .9 & .1 & 0 \\ .1 & .8 & .1 \\ .9 & .1 & 0 \end{pmatrix}$



- HMM for character 'B':

Transition probabilities: $\{a_{ij}\} = \begin{pmatrix} .8 & .2 & 0 \\ 0 & .8 & .2 \\ 0 & 0 & 1 \end{pmatrix}$

Observation probabilities: $\{b_{jk}\} = \begin{pmatrix} .9 & .1 & 0 \\ 0 & .2 & .8 \\ .6 & .4 & 0 \end{pmatrix}$



Exercise: character recognition with HMM(2/3)

- Suppose that after character image segmentation the following sequence of island numbers in 4 slices were observed:

$\{1, 3, 2, 1\}$

- *What HMM is more likely to generate this observation sequence, HMM for 'A' or HMM for 'B'?*

Exercise: character recognition with HMM(3/3)

Consider likelihood of generating given observation for each possible sequence of hidden states:

- HMM for character 'A':

Hidden state sequence	Transition probabilities	Observation probabilities
$s_1 \rightarrow s_1 \rightarrow s_2 \rightarrow s_3$	$.8 * .2 * .2$	$* .9 * 0 * .8 * .9 = 0$
$s_1 \rightarrow s_2 \rightarrow s_2 \rightarrow s_3$	$.2 * .8 * .2$	$* .9 * .1 * .8 * .9 = 0.0020736$
$s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_3$	$.2 * .2 * 1$	$* .9 * .1 * .1 * .9 = 0.000324$
Total = 0.0023976		

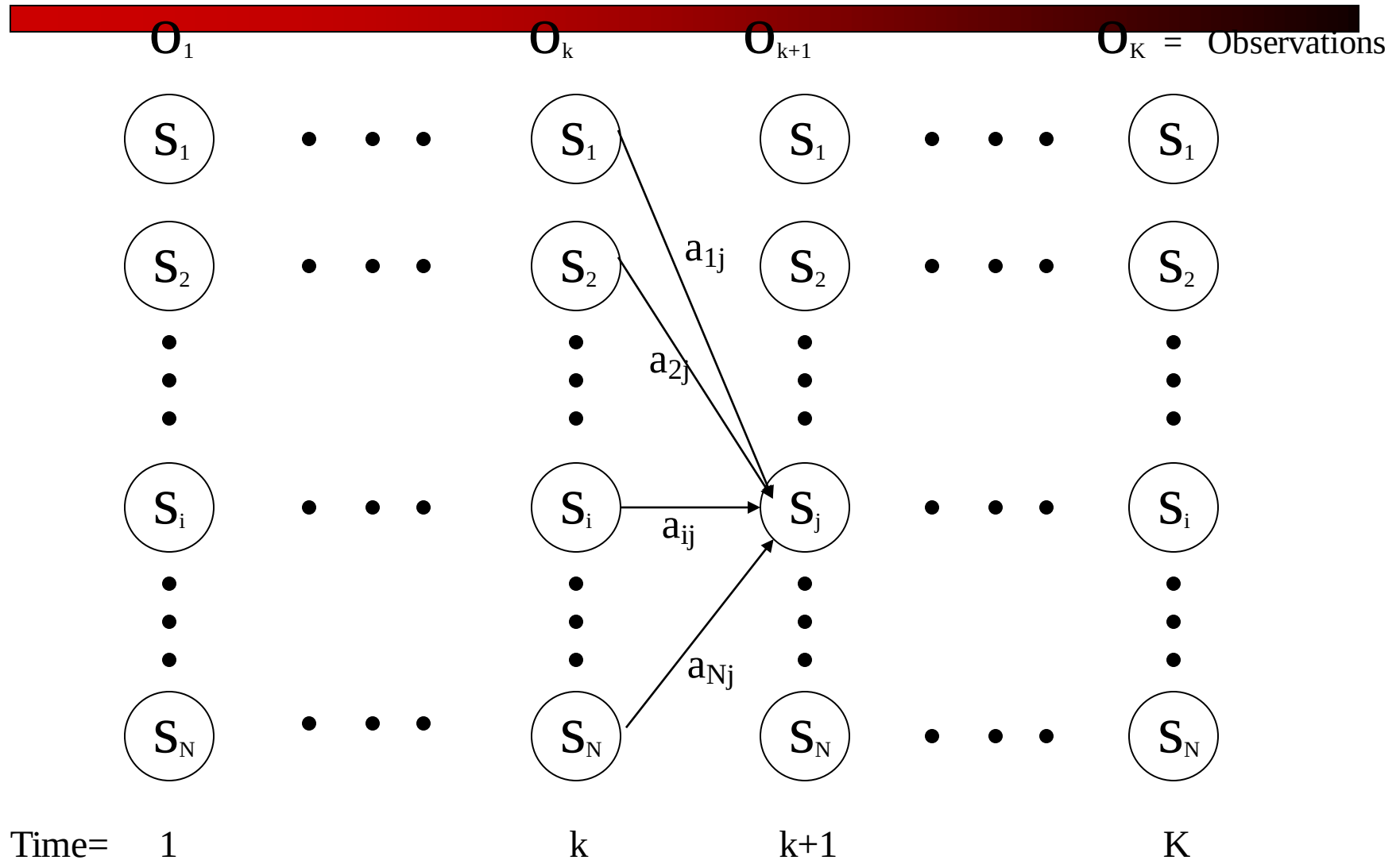
- HMM for character 'B':

Hidden state sequence	Transition probabilities	Observation probabilities
$s_1 \rightarrow s_1 \rightarrow s_2 \rightarrow s_3$	$.8 * .2 * .2$	$* .9 * 0 * .2 * .6 = 0$
$s_1 \rightarrow s_2 \rightarrow s_2 \rightarrow s_3$	$.2 * .8 * .2$	$* .9 * .8 * .2 * .6 = 0.0027648$
$s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_3$	$.2 * .2 * 1$	$* .9 * .8 * .4 * .6 = 0.006912$
Total = 0.0096768		

Evaluation Problem

- **Evaluation problem.** Given the HMM $M=(A, B, \pi)$ and the observation sequence $O=o_1 o_2 \dots o_K$, calculate the probability that model M has generated sequence O
- Trying to find probability of observations $O=o_1 o_2 \dots o_K$ by means of considering all hidden state sequences (*as was done in example*) is impractical:
 - N^K hidden state sequences-exponential complexity
- Use **Forward-Backward HMM algorithms** for efficient calculations
- Define the forward variable $\alpha_k(i)$ as the joint probability of the partial observation sequence $o_1 o_2 \dots o_k$ and the hidden state s_i at time k : $\alpha_k(i)=P(o_1 o_2 \dots o_k, q_k=s_i)$

Trellis representation of an HMM



Forward recursion for HMM

- Initialization:

$$\alpha_1(i) = P(o_1, q_1 = s_i) = \pi_i b_i(o_1), \quad 1 \leq i \leq N$$

- Forward recursion:

$$\begin{aligned} \alpha_{k+1}(j) &= P(o_1 o_2 \dots o_{k+1}, q_{k+1} = s_j) = \\ &\sum_i P(o_1 o_2 \dots o_{k+1}, q_k = s_i, q_{k+1} = s_j) = \\ &\sum_i P(o_1 o_2 \dots o_k, q_k = s_i) a_{ij} b_j(o_{k+1}) = \\ &[\sum_i \alpha_k(i) a_{ij}] b_j(o_{k+1}), \quad 1 \leq j \leq N, \quad 1 \leq k \leq K-1 \end{aligned}$$

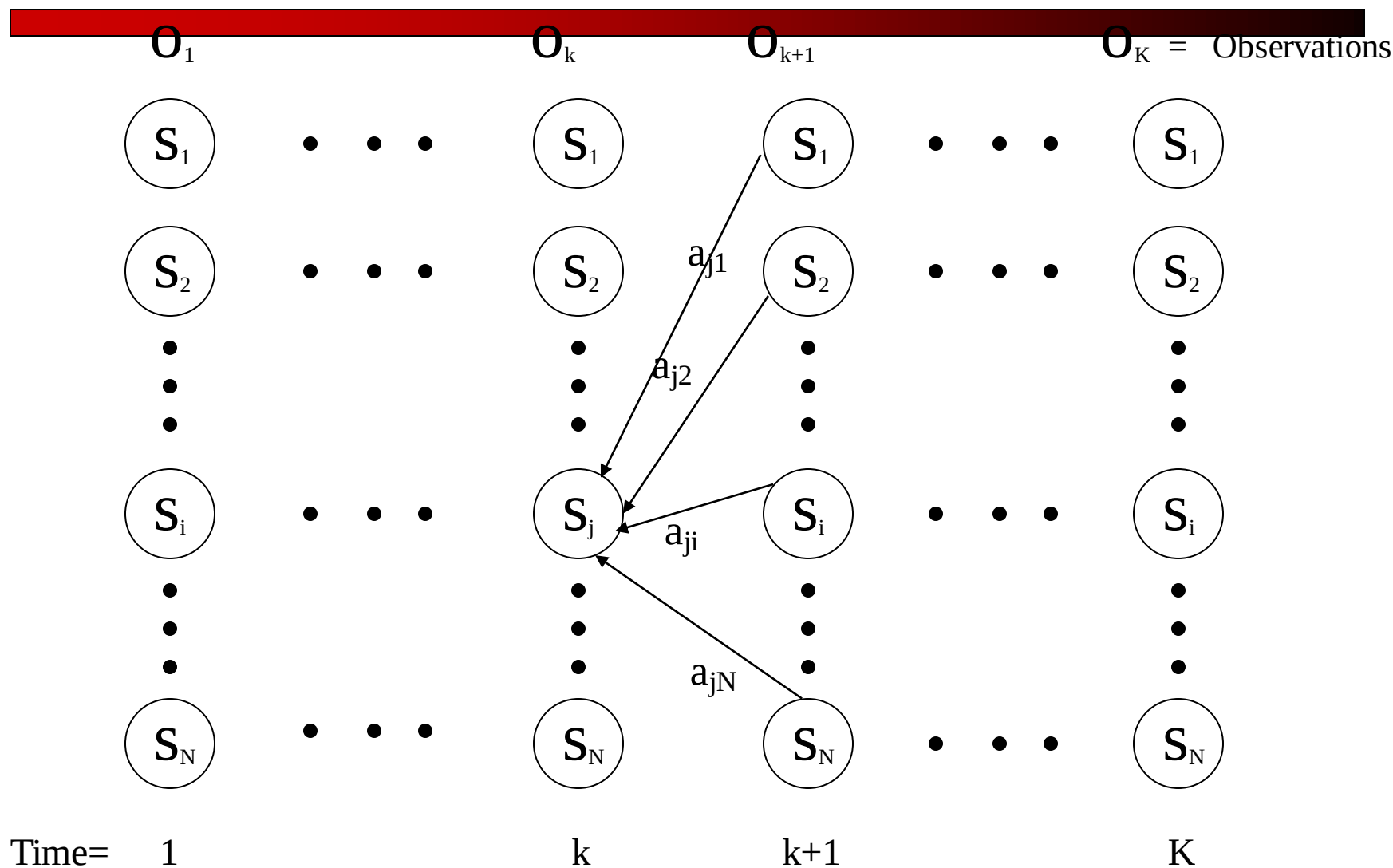
- Termination:

$$P(o_1 o_2 \dots o_K) = \sum_i P(o_1 o_2 \dots o_K, q_K = s_i) = \sum_i \alpha_K(i)$$

- Complexity :

$N^2 (K - 1) + N$ operations ? (*Check*)

Trellis representation of an HMM



Backward recursion for HMM

• Define the backward variable $\beta_k(i)$ as the joint probability of the partial observation sequence $o_{k+1} o_{k+2} \dots o_K$ given that the hidden state at time k is s_i : $\beta_k(j) = P(o_{k+1} o_{k+2} \dots o_K \mid q_k = s_j)$

• Initialization:

$$\beta_K(i) = 1, \quad 1 \leq i \leq N.$$

• Backward recursion:

$$\begin{aligned} \beta_k(j) &= P(o_{k+1} o_{k+2} \dots o_K \mid q_k = s_j) = \\ &\sum_i P(o_{k+1} o_{k+2} \dots o_K, q_{k+1} = s_i \mid q_k = s_j) = \\ &\sum_i P(o_{k+2} o_{k+3} \dots o_K \mid q_{k+1} = s_i) a_{ji} b_i(o_{k+1}) = \\ &\sum_i \beta_{k+1}(i) a_{ji} b_i(o_{k+1}), \quad 1 \leq j \leq N, \quad 1 \leq k \leq K-1. \end{aligned}$$

• Termination:

$$\begin{aligned} P(o_1 o_2 \dots o_K) &= \sum_i P(o_1 o_2 \dots o_K, q_1 = s_i) = \\ &\sum_i P(o_1 o_2 \dots o_K \mid q_1 = s_i) P(q_1 = s_i) = \sum_i \beta_1(i) b_i(o_1) \pi_i \end{aligned}$$

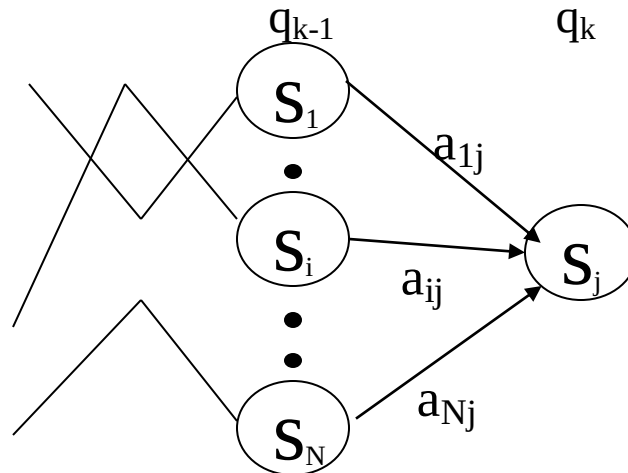
Decoding Problem

- **Decoding problem.** Given the HMM $M=(A, B, \pi)$ and the observation sequence $O=o_1 o_2 \dots o_K$, calculate the most likely sequence of hidden states s_i that produced this observation sequence
- We want to find the state sequence $Q= q_1 \dots q_K$ which maximizes $P(Q | o_1 o_2 \dots o_K)$, or equivalently $P(Q, o_1 o_2 \dots o_K)$
- Brute force consideration of all paths takes exponential time. Use efficient **Viterbi algorithm** instead
- Define variable $\delta_k(i)$ as the maximum probability of producing observation sequence $o_1 o_2 \dots o_k$ when moving along any hidden state sequence $q_1 \dots q_{k-1}$ and getting into $q_k = s_i$.
$$\delta_k(i) = \max P(q_1 \dots q_{k-1}, q_k = s_i, o_1 o_2 \dots o_k)$$
where max is taken over all possible paths $q_1 \dots q_{k-1}$

Viterbi algorithm (1/2)

- **General idea:**

if best path ending in $q_k = s_j$ goes through $q_{k-1} = s_i$ then it should coincide with best path ending in $q_{k-1} = s_i$



- $\delta_k(j) = \max P(q_1 \dots q_{k-1}, q_k = s_j, o_1 o_2 \dots o_k) = \max_i [a_{ij} b_j(o_k) \max P(q_1 \dots q_{k-1} = s_i, o_1 o_2 \dots o_{k-1})]$

- To backtrack best path keep info that predecessor of s_j was s_i .

Viterbi algorithm (2/2)

- Initialization:

$$\delta_1(i) = \max P(q_1 = s_i, o_1) = \pi_i b_i(o_1), 1 \leq i \leq N$$

- Forward recursion:

$$\begin{aligned} \delta_k(j) &= \max P(q_1 \dots q_{k-1}, q_k = s_j, o_1 o_2 \dots o_k) = \\ &= \max_i [a_{ij} b_j(o_k) \max P(q_1 \dots q_{k-1} = s_i, o_1 o_2 \dots o_{k-1})] = \\ &= \max_i [a_{ij} b_j(o_k) \delta_{k-1}(i)], \quad 1 \leq j \leq N, 2 \leq k \leq K. \end{aligned}$$

- Termination: choose best path ending at time K

$$\max_i [\delta_K(i)]$$

- Backtrack the best path

This algorithm is similar to the forward recursion of evaluation problem, with Σ replaced by max and additional backtracking

Learning problem (1/2)

- **Learning problem.** Given some training observation sequences $O=o_1 o_2 \dots o_K$ and general structure of HMM (numbers of hidden and visible states), determine HMM parameters $M=(A, B, \pi)$ that best fit training data, i.e. maximizes $P(O | M)$
- Finding optimal parameter values- *very difficult* (no existing algorithm)
- Use *iterative expectation-maximization algorithm* to find local maximum of $P(O | M)$ -**Baum-Welch algorithm**

Learning problem (2/2)

- If training data has information about sequence of hidden states (as in word recognition example), then use maximum likelihood estimation of parameters:

$$a_{ij} = P(s_i | s_j) = \frac{\text{Number of transitions from state } s_j \text{ to state } s_i}{\text{Number of transitions out of state } s_j}$$

$$b_i(v_m) = P(v_m | s_i) = \frac{\text{Number of times observation } v_m \text{ occurs in state } s_i}{\text{Number of observation occurrences in state } s_i}$$

Baum-Welch algorithm



$$a_{ij} = P(s_i | s_j) = \frac{\text{Expected number of transitions from state } s_j \text{ to state } s_i}{\text{Expected number of transitions out of state } s_j}$$

$$b_i(v_m) = P(v_m | s_i) = \frac{\text{Expected number of times observation } v_m \text{ occurs in state } s_i}{\text{Expected number of total observations in state } s_i}$$

$$\pi_i = P(s_i) = \text{Expected frequency in state } s_i \text{ at time } k=1.$$

Disadvantage of HMMs (1)

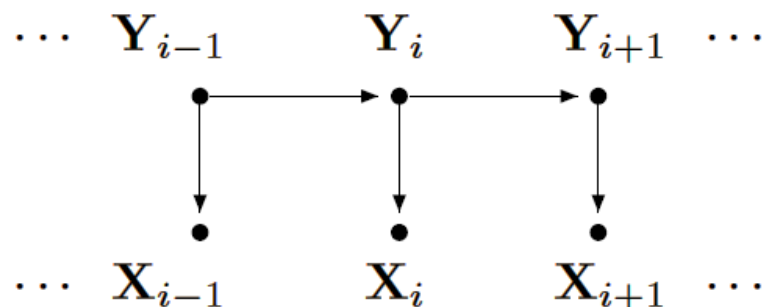
- No Rich Feature Information
 - Rich information are required
 - When x_k is complex
 - When data of x_k is sparse
- Example: Part-of-Speech (PoS) Tagging
 - How to evaluate $P(w_k/t_k)$ for unknown words w_k ?
 - Useful features
 - Suffix, e.g., -ed, -tion, -ing, etc.
 - Capitalization
- Generative Model
 - Parameter estimation: maximize the joint likelihood of training examples

$$\sum_{(\mathbf{x}, \mathbf{y}) \in T} \log_2 P(\mathbf{X} = \mathbf{x}, \mathbf{Y} = \mathbf{y})$$

Generative Models

- Hidden Markov models (HMMs) and stochastic grammars
 - Assign a joint probability to paired observation and label sequences
 - The parameters typically trained to maximize the joint likelihood of train examples

Standard tool is the hidden Markov Model (HMM).



$$P(\mathbf{X}, \mathbf{Y}) = \prod_i P(\mathbf{X}_i | \mathbf{Y}_i) P(\mathbf{Y}_i | \mathbf{Y}_{i-1})$$



Thank you